Jose Fernandes

Rust Systems Software Engineer

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Professional Summary

Rust-focused Systems Software Engineer with 4 years of experience building efficient, secure, and maintainable distributed systems and performance-critical software. Track record of engineering-lead role across multi-disciplinary engineering teams to deliver measurable production outcomes in other domains such as robotics, artificial intelligence, computer vision and cloud data engineering. Strong interest in type-driven development and formal verification methods as enablers of safety and reliability in production systems.

Technical Expertise

- **Rust**: async/await, tokio (channels, select, cancellation tokens, task trackers), sqlx, testing, smart-contract integration (Sui, Cosmos SDKs);
- **Languages**: Rust, Haskell, Ocaml, Python, Java, C/C++;
- **Systems**: Linux, Nix, Docker, RaspberryPi GPIO, Nvidia Jetson, WebAssembly, HTTP/REST, WebSockets, Protobuf, gRPC, OpenTelemetry, Grafana;
- **Databases**: PostgreSQL, Azure SQL, NoSQL;
- **Cloud**: Azure (Synapse, Functions, Event Hub), AWS (S3, EC2), Terraform;

Professional Experience

(052025 -) Rust Software Engineer **Enoda Ltd** **Edinburgh, UK**

Co-engineering a new energy aggregator platform, as part of a 2-person Rust team, a market- and hardware-agnostic SDK, built on a composable, pluggable actor-model architecture for real-time energy bidding. The platform is designed to be bootstrapped and deployed, in parallel, across multiple TSOs (Transmission Service Operator) in different jurisdictions. Given the platform's long production timeline, the team's primary goal (and key metric) this year is reaching integration testing with an energy market.

- Proposed, and then co-designed an actor-model architecture to unify implementation, business/functional requirements and documentation, enabling effective tracking of current and future requirements/specification changes;
- Co-designed and implemented core framework components in Rust on tokio and PostgreSQL (sqlx), using a clean separation of concerns across framework layers;
- Engineered robust I/O-bound concurrency model, leveraging tokio primitives, with fault-tolerant supervision of running instances and their internal resources;
- Set up comprehensive unit testing coverage across framework's primitives as well as integration testing (and technical documentation) tightly aligned with functional requirements;
- Installed a monitoring and alerting layer, leveraging OpenTelemetry and Grafana, to trace application and business operations, and handle certain kinds of runtime errors;
- Reached integration testing stage with an EU-based energy market, the team's key delivery milestone for the year.

(112022 - 052025) Software Engineer **Robotarium** **Edinburgh, UK**

Worked across a portfolio of industrial production-driven AI and Robotics R&D projects for major UK industrial clients, progressing to project lead engineer;

- Developed a turn-taking detection algorithm for an automotive conversational application which improved conversation completion rates by 13%, reported by a car manufacturer's internal experiments;
- Integrated a real-time robotic bottle defect QA system improving issue detection by an estimate of 30% over trained human inspectors and saving £200k/year, according to the client's reports;
- Contributed to development of cloud data platform, for oil rig automated inspections, leveraging Azure Synapse, Terraform-automated infrastructure, and a WebAssembly Blazor frontend;
- Led a team to deliver a computer-vision garment defect detection system (using AWS) achieving 87% IoU, with the client forecasting overall productivity improvement by 2%.

Education

Masters - MEng Software Engineering, Heriot-Watt University (2018 - 2023)

Awarded **Distinction**.

Relevant Projects

Football Analytics Engine

Proprietary Haskell application (nixified infrastructure) that ingests in-match football event streams and derives performance metrics distinct from standard commercial offerings. Personal entrepreneurial project targeting eventual commercial launch.

Open-Source Contributor P2PRC

Contributed to software architecture and system integration planning. Developed and maintained the Haskell language API layer. Managed builds, development environments, and deployments with Nix.

Docker Development & Deployment environment

Developed a lightweight Bash framework that standardised Docker development and deployment workflows across multiple contracted projects, while at Robotarium; included Nvidia runtime and X11 desktop application support.

Languages

- **English**: full professional proficiency;
- **Portuguese**: native;
- **Spanish**: professional proficiency;